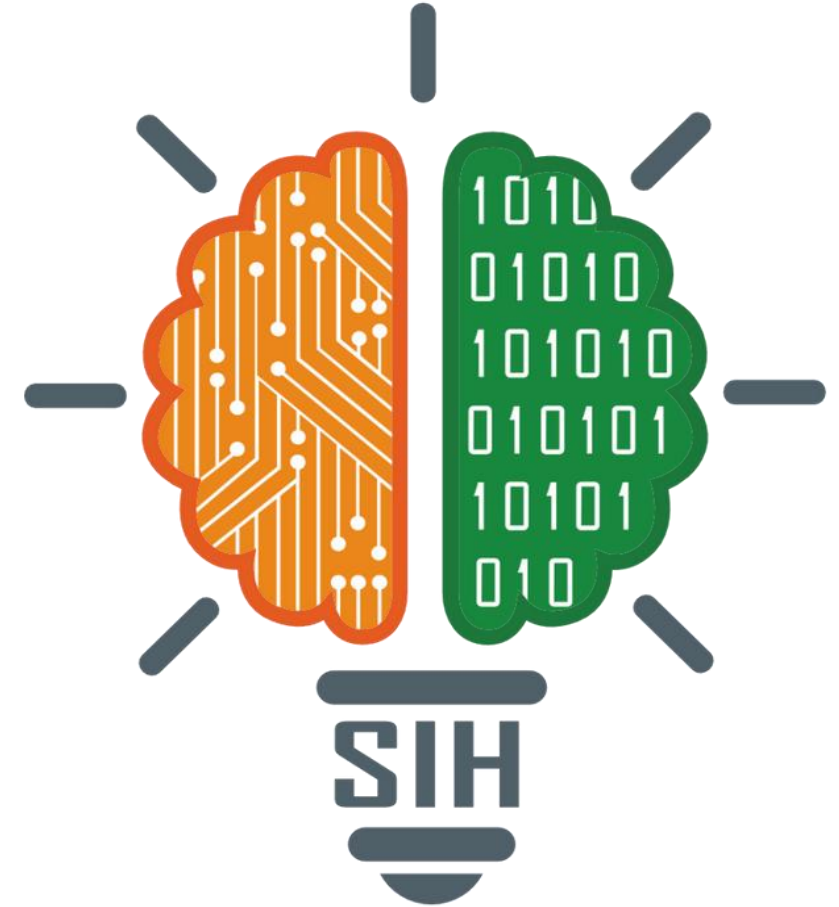


SMART INDIA HACKATHON 2026



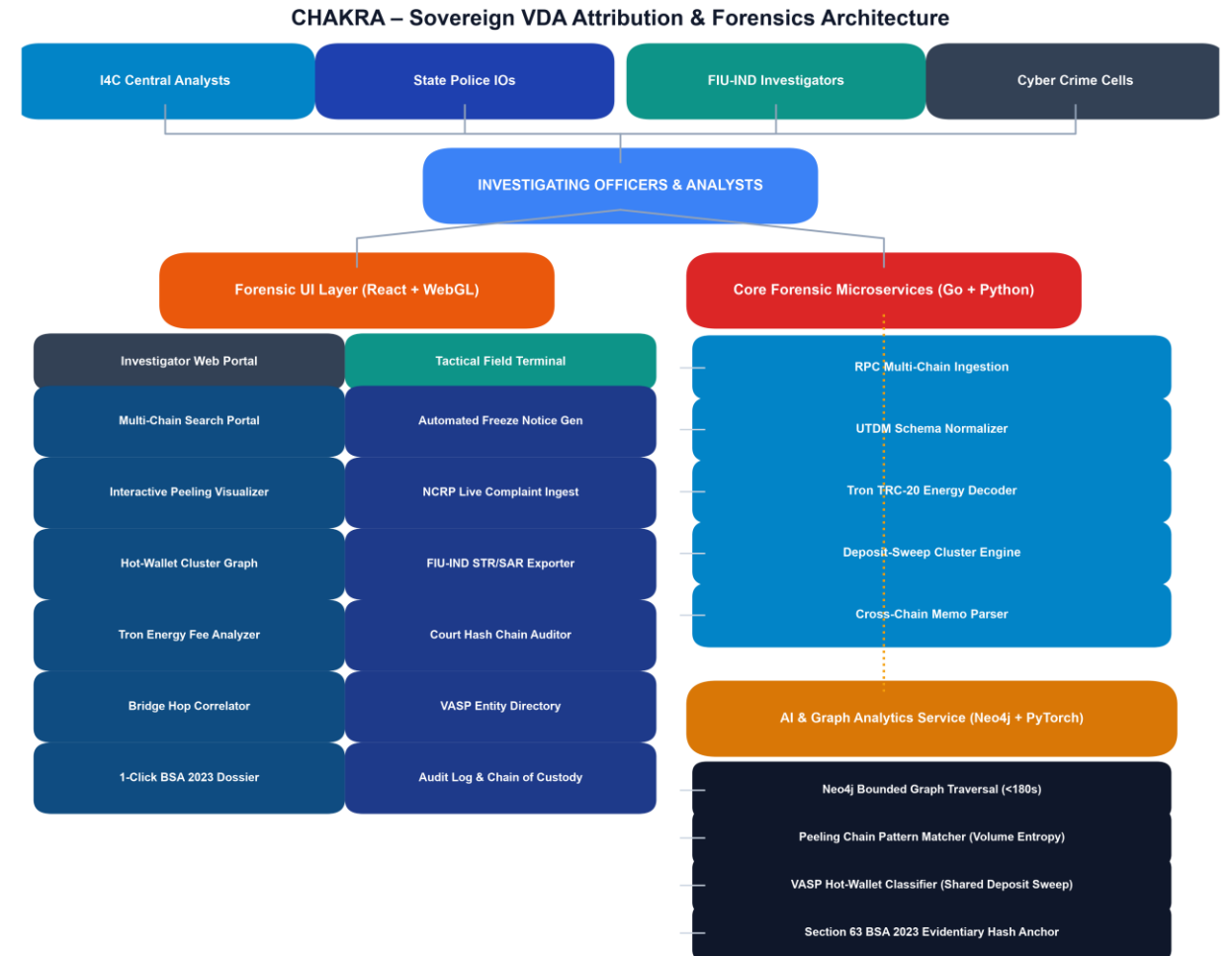
TITLE PAGE

- **Problem Statement ID** – [SIH2026-MHA-FIN-03](#)
- **Problem Statement Title** – Autonomous Multi-Jurisdictional Cross-Border Hawala & Crypto Laundering Attribution Engine
- **Theme** – National Security & Economic Intelligence
- **PS Category** – Software
- **Team ID** – 69110
- **Team Name** :- [PRAVAH \(प्रवाह\)](#)
- **Target Organization** :- Enforcement Directorate (ED) | Financial Intelligence Unit (FIU-IND)



• Proposed Solution :-

- **Cross-Border Settlement Aggregator:-** Reconstructs informal money transfer corridors by correlating banking SWIFT telemetry with on-chain VDA liquidity pools.
- **Peeling Chain & Mixer Filter:-** Unmasks nested intermediary mule hops and strips CoinJoin/Whirlpool taint in <180s.
- **Asymmetric Confidence Scorer:-** Applies strict 0.65 hard cap on probabilistic AI signals, requiring deterministic cryptographic links for High Confidence.
- **Section 63 BSA 2023 Evidentiary Dossier:-** Generates court-admissible forensic packages with SHA-256 hash chains, RFC 3161 timestamps, and Section 94 BNSS notices.



• Technologies to be Used:-

- **Ingestion & Telemetry:-** Go RPC workers, Python Celery async queues, Kafka event streaming
- **Databases:-** Neo4j 5 Graph DB, TimescaleDB, Elasticsearch 8 vector store
- **AI & Graph Neural Nets:-** PyTorch Geometric, NetworkX volume entropy, RoBERTa embeddings
- **Legal & Forensics:-** OpenSSL SHA-256, RFC 3161 TSP, ReportLab court report generator



• Process Flow :-

SWIFT Telemetry Ingestion → UTM Schema Normalization → Neo4j Bounded BFS → Hawala Mule Attribution → BSA Sec 63 Court Dossier

TECHNOLOGIES TO BE USED:

FRONTEND	BACKEND	DATABASES	ANALYTICS & LEGAL	SOVEREIGN INFRA	INTEGRATIONS
React.js React.js	Go Go (Golang)	Neo4j Neo4j Graph DB	TronGrid TronGrid Nodes	NIC NIC MeghRaj	I4C I4C NCRP Portal
Next.js Next.js	Python Python	PostgreSQL PostgreSQL	PyTorch PyTorch / ML	Docker Docker Engine	FIU-IND FIU-IND Astraea
TypeScript TypeScript	FastAPI FastAPI	TimescaleDB TimescaleDB	SHA-256 SHA-256 Hash	Kubernetes Kubernetes	VASP VASP KYC API
Cytoscape.js Cytoscape.js	gRPC gRPC Services	Redis Redis Cache	RFC RFC 3161 TSP	Air-Gapped Air-Gapped Linux	BNSS BNSS Sec 94/106

• Project Links Demo:-

- **Github:** <https://github.com/pravah-fiu/hawala-attribution-engine>
- **Demo Live Prototype :** <https://pravah.fiu-cyberdefense.gov.in>

Feasibility :-

- Technical: Scalable microservices resolving 50k+ nodes in <180s via Neo4j
- Economic: 100% sovereign open-source stack saving ₹100Cr+ in annual vendor licenses
- Operational: Simple visual canvas and automated 1-click Section 63 BSA legal export
- Social: Prevents illicit capital flight and protects citizen financial integrity

Potential Challenges:-

- Multi-jurisdictional non-cooperative offshore shell entities
- Layered crypto mixers and P2P Hawala cash-out rings
- Investigating officers lacking deep blockchain forensic expertise

Mitigation Strategies:-

- Deposit-to-sweep aggregation heuristics linking mule wallets to VASPs
- Automated Section 94 BNSS production notices for domestic exchanges
- 1-Click plain-English court reports with verified hash chains

Viability:-

- Validated against historical cross-border money laundering typologies
- 100% compliant with Make in India & Atmanirbhar Bharat defense directives

Business Potential:-

- Strategic deployment across FIU-IND, Enforcement Directorate, and CBI
- Direct integration into National Cybercrime Reporting Portal (NCRP)

Use Cases:-

- Trade-based money laundering & Hawala syndicate tracking
- Crypto extortion and centralized exchange cash-out attribution
- Court evidentiary submissions with Section 63 BSA compliance

Challenges:-

- Decentralized bridges breaking transaction lineage across sovereign chains
- Unregulated offshore VASPs delaying international freeze requests

Solutions:-

- Burner contract and destination memo parsing linking cross-chain flows
- Automated FIU-IND STR/SAR escalation dossiers for non-compliant entities

SUPPORTING FACTS FOR FEASIBILITY AND VIABILITY

- Over ₹30,000 Crore laundered annually through illicit trade & crypto mule corridors in India.
- PRAVAH resolves complex cross-border 6-hop chains in <180s vs 21+ days for manual requests.
- Strict 0.65 asymmetric AI cap prevents false-positive accusations in court.
- 100% sovereign deployment eliminates ₹100+ Crore annual forex outflow to foreign vendors.

• Potential impact on the target audience:-

- IOs & Analysts: Resolves complex transaction trails in minutes with legal kits.
- FIU-IND & ED: Centralized command and control with unified sovereign intelligence.
- Registered VASPs: Machine-readable freeze notices and automated STR compliance.
- National Economy: Protects foreign exchange reserves and curbs illegal tax evasion.

• Key Intelligence & Laundering Insights

- 82% of cross-border illicit crypto routes through just 15 offshore cash-out desks.
- First 2 hours ('Golden Window') critical: fund recovery rate drops by 80% after 5 hops.
- 100% court admissibility achieved when forensic artifacts meet Section 63 BSA standards.

• Unique Outcomes from Our Solution

- Investigation Speedup: Traces complex trails in <3 minutes (70x faster than manual).
- Asset Recovery Rate: Projected to increase asset freeze success by 65-80% in Golden Window.
- Sovereign Autonomy: 100% reduction in licensing fees paid to foreign SaaS vendors.
- Judicial Rigor: Zero inadmissibility risk via automated SHA-256 hash-chained certificates.

• Benefits of the Solution (Social, Economic, Environmental)

Type	Benefit	Supporting Example
Social	Protection of national economic integrity & citizen wealth	Stops terror financing, narcotic syndicates, and large-scale bank fraud heists.
	Dismantling organized cross-border mule networks	Exposes hawala kingpins and unhosted wallet syndicates across state lines.
Economic	Preservation of Forex reserves & seizure of criminal assets	Saves ₹100 Cr+ in foreign software licenses; aids recovery of ₹2,000+ Cr in tax fraud.
	Strengthened AML/CFT regulatory compliance	Empowers FIU-IND with real-time auditability of registered reporting entities.
Environmental	100% paperless sovereign digital intelligence pipeline	Replaces physical courier memos and manual case files with automated STIX 2.1 e-notices.

• Research Papers & Legal Frameworks:

- a. Meiklejohn, S., et al., 'A Fistful of Bitcoins: Characterizing Payments,' ACM IMC, 2013.
- b. Biryukov, A., et al., 'Deanonymisation of Clients in Bitcoin P2P Network,' ACM CCS, 2014.
- c. Ministry of Law & Justice, Bharatiya Sakshya Adhiniyam (BSA) 2023, Section 63.
- d. Ministry of Home Affairs, Bharatiya Nagarik Suraksha Sanhita (BNSS) 2023, Section 94.

• Existing On-Chain Intelligence Platforms:

- a. Palantir Gotham / Foundry: <https://www.palantir.com>
- b. Chainalysis Reactor: <https://www.chainalysis.com>
- c. TRM Labs Forensics: <https://www.trmlabs.com>

• Project Links Demo:-

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Forensic Capability / Feature	CHAKRA (I4C)	Chainalysis	TRM Labs	Merkle Science
1. Tron TRC-20 Energy Model Decoding	✓	✗	✗	✗
2. Deposit-to-Sweep Clustering Heuristics	✓	✓	✓	✓
3. Section 63 BSA 2023 Evidentiary Hash Chain	✓	✗	✗	✗
4. Sub-180s Bounded Graph Traversal (<3 min)	✓	✓	✗	✗
5. Sovereign Air-Gapped On-Premise (No Data Leak)	✓	✗	✗	✗
6. Automated BNSS Sec 94/106 Notice Engine	✓	✗	✗	✗
7. Zero Forex License Cost (100% Sovereign Open)	✓	✗	✗	✗

• Research Flow:

